EE3001: Solid State Devices
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Assignment #10

Try yourself to solve the problems with detailed steps. Take a limited time quiz to test your understanding on related concepts.

Problems will be discussed in class. You need to contact TAs or teacher if you are unable to do anything useful by yourself or with the help of your friends.

1. (1 point) In a metal/SiO_2/p-Si MOS capacitor, the SiO_2 layer thickness is t_{ox} and the doping concentration of the p-type substrate is N_A. The threshold voltage of the MOS capacitor will definitely increase if (pick the correct option)

   (A) t_{ox} is decreased and N_A is increased

   (B) t_{ox} is increased and N_A is decreased

   (C) both t_{ox} and N_A are decreased

   (D) both t_{ox} and N_A are increased

2. (8 points) (a) Assuming that the flatband voltage (V_{FB}) is zero, find the threshold voltage in the ideal case for a MOS capacitor with a substrate doping concentration N_A =10^{15} $/cm^3$ and the gate oxide thickness t_{ox} = 100 nm.

   (b) Repeat the same for different substrate doping concentrations (i) $N_A = 10^{14}$ $/cm^3$; (ii) $N_A = 10^{16}$ $/cm^3$; (iii) $N_A = 10^{17}$ $/cm^3$

   (c) If the above MOS capacitors have Aluminium gate, find the threshold voltage in each case considering the difference in work function. [Electron affinity of silicon is 4.05 eV and work function of Aluminium is 4.1 eV]

   (d) Repeat the same for MOS capacitors with (i) n^+ polysilicon gate and (ii) p^+ polysilicon gate

3. (10 points) Fig. 1 shows the C-V characteristics of a metal-oxide-silicon (MOS) capacitor, where the doping concentration in silicon is 2×10^{16} $/cm^3$ and the metal has a work function ($q\phi_m$) of 4.1 eV. For silicon, the electron affinity ($q\chi_s$) is 4.05 eV. Assume that there are no charges in the oxide.

   (a) What is the type (n-type/p-type) of silicon? Give reasons.

   (b) At what frequency (high frequency/low frequency) was the C-V characteristics measured? Give reasons.

   (c) If the area of the capacitor is 1 mm^2, what is the gate oxide thickness (t_{ox})?

   (d) What is the flat-band voltage (V_{FB})?

   (e) What is the threshold voltage (V_{th})?

   (f) What is the minimum capacitance (C_{min}) in the C-V characteristics?

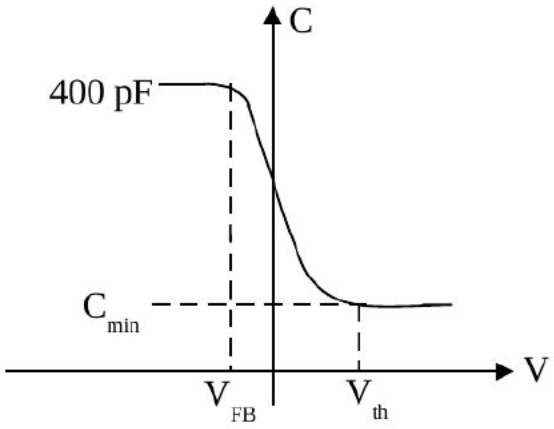


Figure 1: Bias dependent capacitance in MOS device.

4. (4 points) A long n-channel MOSFET has the following SPICE model parameters: W (width) = 10 μm, L (length) = 1.0 μm, V_{TO}(zero bias threshold voltage) = 1 V, $\mu_n C_{ox}$ (transconductance parameter) = 5×10^{-4} A/V^2, GAMMA (body effect parameter) = 0.5 $V^{0.5}$, PHI (= $2\phi_B$) = 0.9 V, T_{OX} (gate oxide thickness) = 3 nm, T (temperature) = 300 K.

   (a) What is the drain current (I_D) calculated for this MOSFET model for V_{GS} = 2.1 V, V_{DS}= 1 V and V_{BS} = 0 V from SPICE level 1 model?

   (b) What is the corresponding values when V_{GS} = 2.1 V, V_{DS} = 1 V and V_{BS} = -1 V?

5. (6 points) An n-channel MOSFET has the following specifications: The channel width (W) = 5 μm, The channel length (L) = 1 μm, The oxide thickness (t_{ox}) = 50 nm, electron mobility = 800 cm^2/V.sec, threshold voltage (V_{th}) = 1 V.

   Find the drain current flowing through the device for the following conditions using SPICE Level 1 and Level 3 models (assume α = 1.2):

   (i) V_{GS} = 2 V, V_{DS} = 0.5 V

   (ii) V_{GS} = 2 V = V_{DS}

6. (5 points) An n-channel MOSFET has a threshold voltage (V_{th}) of 0.5 V and a sub-threshold slope of 100 mV/decade. If the drain current (I_D) of this MOSFET obtained from simulations with SPICE Level 3 model at V_{GS} = 1 V and V_DS = 2 V is 1 mA, what is the value of I_D from the same model when V_{GS} = 0.25 V and V_{DS} = 2 V. Assume T = 300 K and V_T = 26 mV.